

Spacetime and spacetime worms

PHIL UA-78 Metaphysics

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B-theorists accept, and A-theorists deny:

Fact Eternalism

Every state of affairs that ever obtains, always obtains.

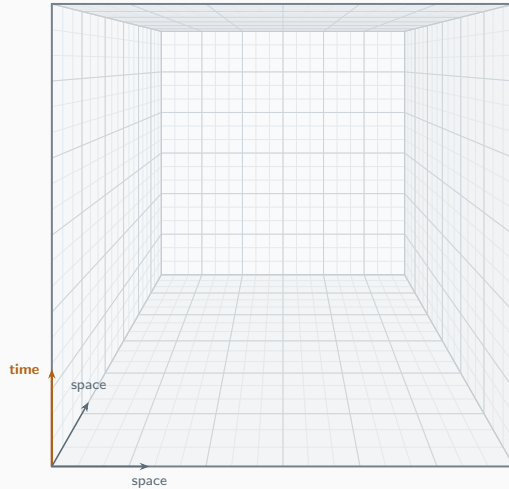
The world as a four-dimensional manifold

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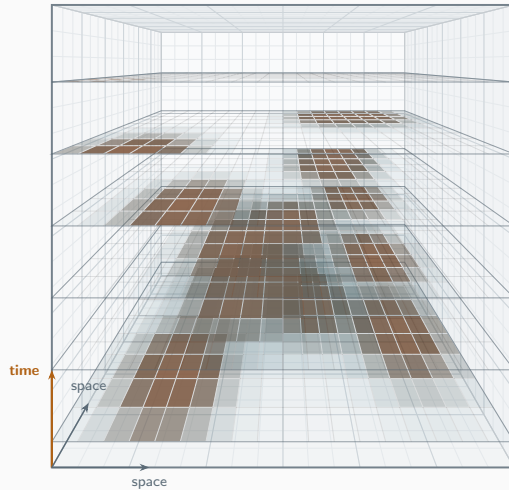
There are points. They stand in geometric relations that give them the structure of a four-dimensional manifold. Maybe they also have some other fundamental properties and relations, which can be described in terms of one or more physical *fields*. Maybe there are, in addition, *particles*, each of which *occupies* a certain collection of points (its ‘worldline’). *And that’s all!*

- ▶ The purest version of the view does *not* add particles as something over and above spacetime—instead, particles are just certain “patterns” in the fields. Focus on this for now.
- ▶ A variant of the view gets rid of points and instead makes the fundamental elements 4d *regions*, understood as “gunky”: each has parts smaller than itself. But physics is much easier to formulate when we have points.

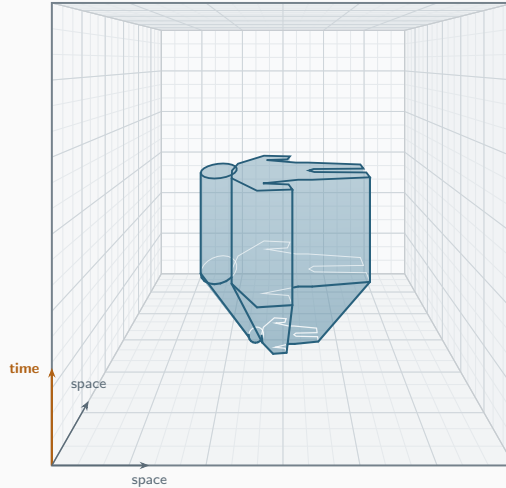
The manifold (one spatial dimension suppressed)



Fields on the manifold



A persisting object is a four-dimensional region



Where does time fit into this picture?

Standard idea: *moments of time* are identical to certain *slices* through the four-dimensional manifold. (We could take these to be *collections* of points).

To answer the question 'How far apart are particles x and y at t ?', you find the distance between the two points where the worldlines of x and y *intersect* t .

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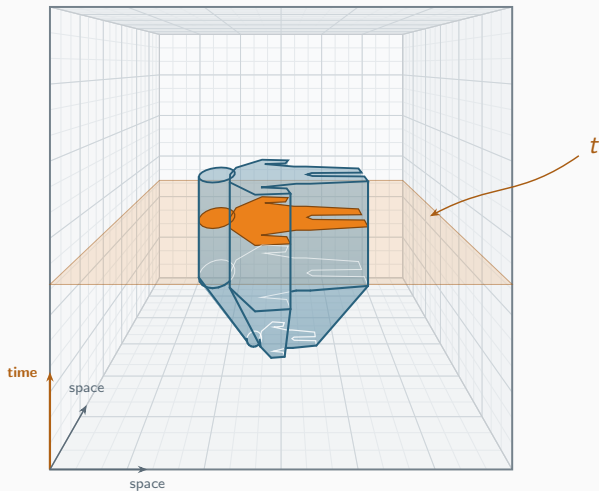
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This fits very naturally with the B-theory, which is already built around an *analogy* between time and space. It is *compatible* with the A-theory, but it is an awkward fit!

A-theorists think that there is something *objectively special* about the present. So if they accept the manifold picture (and identify times in the usual way with slices), they will have to say that over and above the usual physical fields, there is an extra fundamental property not reducible to them—a 'glow of presentness' that (temporarily) lights up a certain slice through the manifold.

A moment of time is a slice through the manifold



Spacetime and absolute velocity

Newton is famous for positing *absolute space*: a collection of points with an eternal Euclidean geometry, that physical objects *occupy* in changing patterns.

Q: Why posit this, rather than just the material bodies with changing distance relations among them—not understood in terms of a relation to something *else*—as some of his predecessors had?

A: to state his famous laws, Newton needed to be able to make sense of *absolute acceleration*. A privileged interpretation of ‘accelerating’ that obeys the law $F = ma$ —not just a matter of accelerating relative to this or that.

Absolute velocity

Unfortunately, the posit of absolute space also forced Newton to think that there is also such a thing as *absolute velocity*: a privileged interpretation of 'moving', not just a matter of moving relative to this or that.

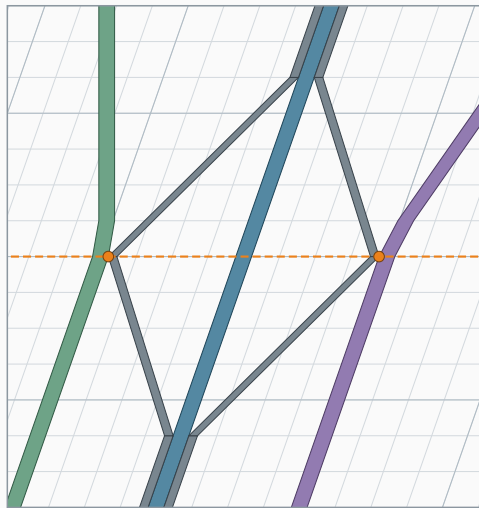
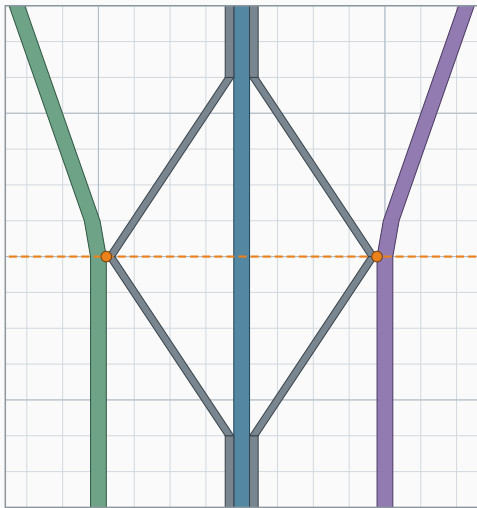
This seems unfortunate, because differences of absolute velocity *make no difference to the physics*. Newton's laws predict that things will look and behave just the same even if we boost everything's velocity in some direction.

There seems to be pressure from Ockham's razor *not* to posit an unmeasurable, physically irrelevant fact about the true absolute velocity of the solar system (or whatever).

Galilean spacetime

Although the idea of spacetime was in fact discovered as a response to relativity theory, we can anachronistically also use it as a solution to Newton's dilemma about absolute velocity. Here's the picture:

- ▶ There is a 4d geometrical manifold of points, which divides in a natural way into 3d hyperplanes ('leaves'), which play the role of instants of time. Each leaf carries Euclidean geometric structure.
- ▶ Material bodies *occupy* 4d paths in this manifold. Newton's laws tell us how these paths curve.
- ▶ While the notion of a *straight line* still applies in the usual way in the 4th dimension, the notion of *perpendicular* does not—there is no sense in which some timelike lines (lines that pass through leaves) are perpendicular to them, and others aren't. Thus, any body whose path is a straight line has an equal claim to be described as "at rest".

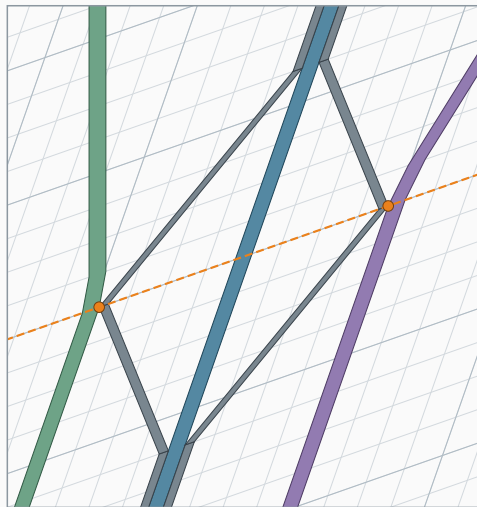
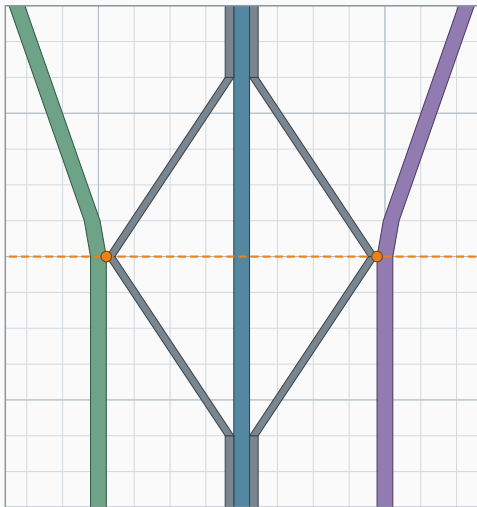


These are two depictions of the same reality.

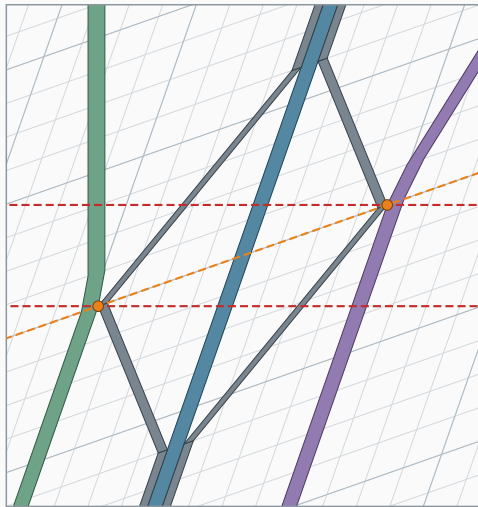
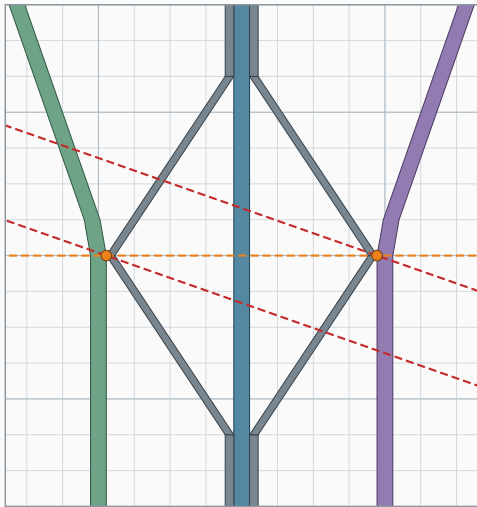
Now we are ready to hear from Einstein. Special Relativity modifies Newton's laws in a way that does appeal to a notion of “perpendicularity” between timelike lines and spacelike hyperplanes.

But the new laws still don't privilege any body whose path is a straight line! Rather, each family of parallel timelike lines is associated with a *different* foliation (division into leaves) of the spacetime: the laws take the same form in any such ‘frame’.

The upshot: if we want to believe that one foliation is ‘special’, we'll also have to go back to Newton's view that one family of parallel timelike lines is ‘special’ (so that there are unmeasurable facts about absolute velocity). If we want to keep the Galilean idea that all non-accelerating observers are ‘on a par’, we'll also have to say that all the corresponding foliations are ‘on a par’.



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The B-theory and context-sensitivity

This is all quite easy to accommodate in a B-theoretic setting.

B-theorists already think that words like 'present', and ordinary present-tensed sentences, are systematically context-sensitive.

And everyone is used to the idea that words like 'moving' (or 'moving in such-and-such direction at such-and-such speed') are context-sensitive: if no-one is accelerating, each of us will find it natural to say 'I am at rest, and everyone else is moving', and no-one is making a mistake.

The B-theorist can just say that the word 'time' is also context-sensitive in a way that correlates with the context-sensitivity of 'moving'.

- And of course this will carry over to lots of temporal sentences that don't use the word 'time', e.g. 'Whenever this light is on, that light is off', or 'Object *a* is always spherical'.

The A-theory and Relativity

The spacetime picture of the world was *already* a very awkward fit for the A-theory. Given that we identify times with 3d hypersurfaces (and it's hard to see what *else* they could be!), the key A-theoretic idea that *the present time is objectively special* requires adding something extra to the picture—a 'spotlight of presentness' or 'degree-of-futurity field', that's totally unlike the usual physical fields.

In the setting of Relativity, it's even worse: if we posit an objectively special 'spotlit' leaf, we'll also have to say that there's an objectively special family of non-accelerating observers—namely those whose worldlines are perpendicular to the special leaf. So we're back with Newton.

Material objects as spacetime regions

Where do material objects fit in?

For simplicity, let's focus on the “purely field-theoretic” version of the manifold picture, that doesn't think of ‘particles’ as something extra in addition to spacetime.

Obviously, ordinary material objects—tables, trees, planets, people...—are not spacetime points! But the manifold picture doesn't have to be understood as implying that ABSOLUTELY EVERYTHING is a point. For example, it seems compatible with the view that there are *regions* made of many points—e.g., the ‘worldlines’ and ‘leaves’ we've been talking about all along.

- At least for our purposes, it is fine to think of regions as *sets* or *collections* or *pluralities* of points—though on some metaphysical views, there are subtle distinctions to be made around here.

Where do material objects fit in?

Here's a natural proposal worth thinking carefully about:

Spatiotemporalism

Every material object is a region of spacetime.

Lewis:

[A person] is a streak through the manifold of space-time, a whole composed of stages located at various times and places.

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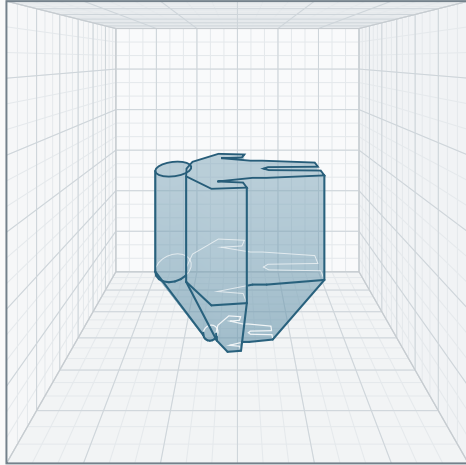
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For example, a typical human being is a 4d region, about 70 years long in the temporal dimension. Its intersection with a leaf perpendicular to its worldline looks like this:

O—|—<.

A material object is a region of spacetime



Understanding spatiotemporalism

This gives a natural program for answering a certain range of questions of the form:
'What is it for x to be F at t ?'

- ▶ What is it for x to be spherical at t ?
 - It's for the intersection of x and t to comprise all and only the points in t within a certain distance of a certain point.
(And similarly for other shape-at- t predicates).
- ▶ What is it for x to have mass m at t ?
 - It's for m to be the result of integrating the "mass density" field over the intersection of x and t . (Likewise for charge and other physical quantities.)
- ▶ What is it for x to exist (be concrete) at t ?
 - It's for the intersection ('meet') of x and t to be nonempty.
- ▶ What is it for x to be at least 40 years old at t ?
 - It's for x to intersect some leaf at least 40 years before t .

The object's stage at t : its intersection with t

